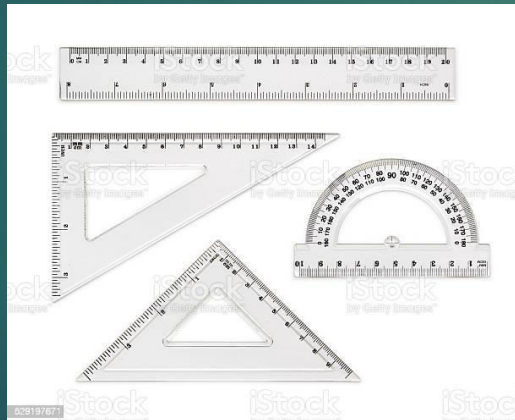


# Experiment (1)

## Measurements and Errors

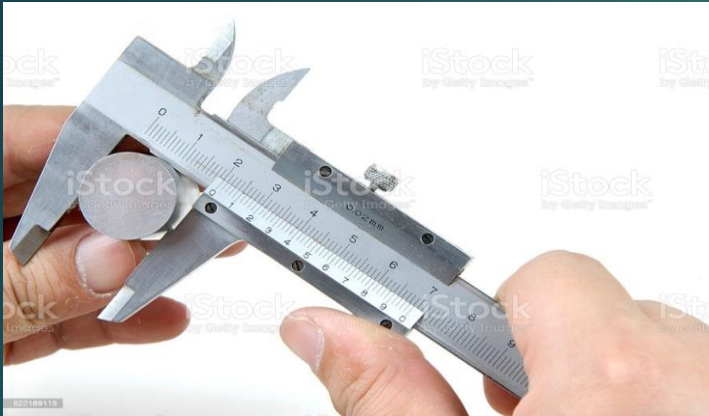


## *Objectives*

1. To learn using Ruler, Vernier Caliber, Micrometer.
2. To estimate the maximum possible error in a measurement.
3. To treat the error calculations to collect the data in tabular form.
4. To estimate the measured quantities and they're errors using the method of residuals.

# *Apparatus*

## Vernier caliber



## Micrometer



## Five different cylinders.



# Physical Quantities

## (الكميات الفيزيائية)

- Measured Quantities (كميات قياسية)

Can be measured directly using suitable instrument  
(Mass, Time, length, Current...)

تقاس بشكل مباشر باستخدام اداة خاصة لهذا القياس مثل ( الكتلة, الزمن, الطول,  
التيار, ...)

- Calculated Quantities (كميات حسابية)

Can be calculated from Measured Quantities (Velocity,  
Volume, Force, Density...)

يمكن حسابها من الكميات القياسية مثل ( السرعة, الحجم, القوة, الكثافة, ..... )

# 1. Errors

## A) Errors in measured Quantities

(يعبر عن الخطأ رياضياً) The error is described by the

- Absoute Error (الخطأ المطلق)

$\Delta X$  = half of the least scale of the insrument (نصف اقل تدريج للاداة)  
 $\Delta X$  = least scale ( If the insrument is digital (اقل تدريج في الادوات الرقمية)

- Relative Error (الخطأ النسبي)

$$\frac{\Delta X}{X}$$

- Percentage Error ((الخطأ النسبي المئوي))

$$\frac{\Delta X}{X} \times 100\%$$

## Rules of calculating errors in the computed quantities

- Multiplying measured quantity by a constant

$$z = a x$$

Where  $a$  is a constant, then the absolute error in  $z$  is given by

$$\Delta z = a \Delta x$$

- Adding and Subtracting measured quantities

$$z = x \pm y$$

$$\Delta z = \Delta x + \Delta y$$

Note that. when measured quantities are **added** or **subtracted**, their errors must also be added (**never subtract**)

- **Multiplying and Dividing measured quantity by a measured quantity**

$$z = xy$$

$$z = \frac{x}{y}$$

When we multiply or dividing measured quantities, we add their relative errors to calculate the relative error of **z**.

$$\frac{\Delta z}{z} = \frac{\Delta x}{x} + \frac{\Delta y}{y}$$

$$\Delta z = z \left[ \frac{\Delta x}{x} + \frac{\Delta y}{y} \right]$$

- **Powers**

$$z = x^n$$

$$\frac{\Delta z}{z} = n \frac{\Delta x}{x}$$

$$\Delta z = z \left[ n \frac{\Delta x}{x} \right]$$



Using the above rules, we can also show that

$$1) \ z = ax + b$$

$$\Delta z = \Delta x$$

$$2) \ z = axy$$

$$\Delta z = z \left[ \frac{\Delta x}{x} + \frac{\Delta y}{y} \right]$$

$$3) \ z = ax^n$$

$$\Delta z = z \left[ n \frac{\Delta x}{x} \right]$$

$$4) \ z = \frac{ax^n}{by^m H^l}$$

$$\Delta z = z \left[ n \frac{\Delta x}{x} + m \frac{\Delta y}{y} + l \frac{\Delta H}{H} \right]$$

Where  $a, b, m$  and  $l$  are constants



Example: A student wants to measure the density of a cylinder. What is the relative error in the density?

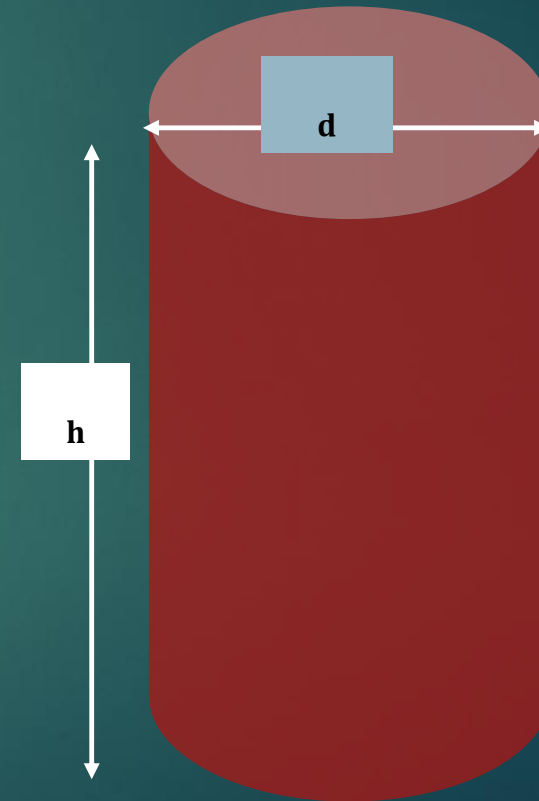
$$\rho = \frac{m}{V}$$

$\rho$ : Density g/cm<sup>3</sup>.  
 $m$ : Mass (g).  
 $V$ : Volume(cm<sup>3</sup>).

$$\rho = \frac{m}{\frac{\pi}{4} d^2 h}$$

$$\frac{\Delta \rho}{\rho} = \frac{\Delta m}{m} + 2 \frac{\Delta d}{d} + \frac{\Delta h}{h}$$

$d$ : Diameter(cm)  
 $h$ : hieght(cm)

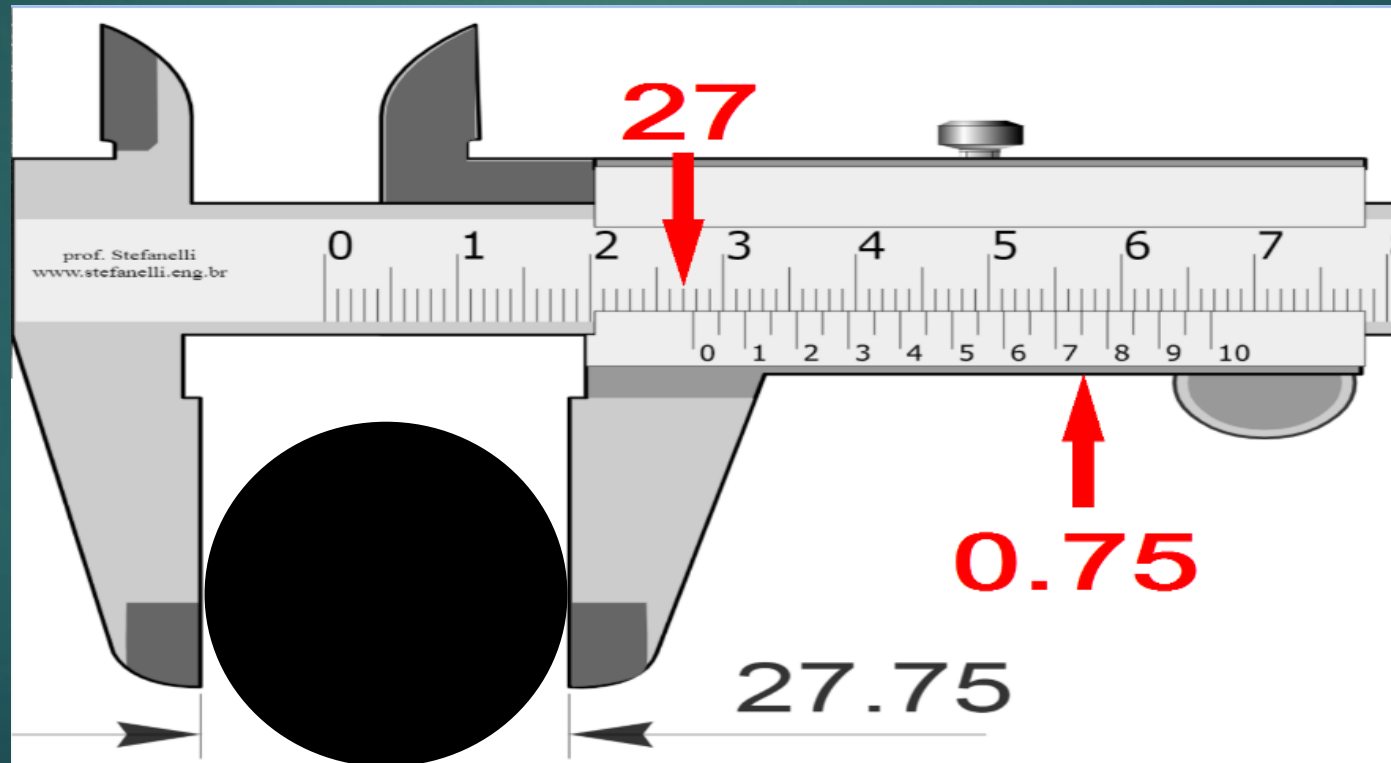


$$\Delta \rho = \rho \left[ \frac{\Delta m}{m} + 2 \frac{\Delta d}{d} + \frac{\Delta h}{h} \right]$$

# The Vernier Caliper

The least scale is 0.05mm. then the absolute error in the reading is 0.025mm

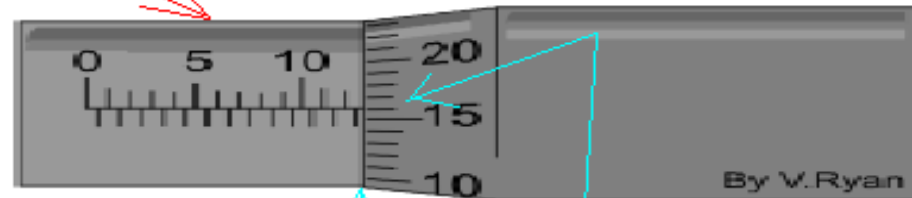
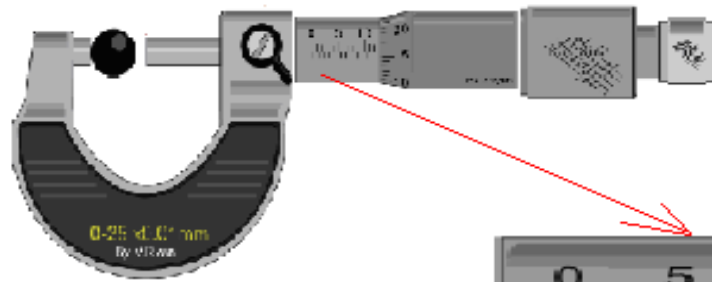
اقل تدريج في هذه الاداة هو 0.05mm لذا يكون الخطأ في هذه الاداة هو 0.025mm



# The Micrometer

The least scale is 0.01mm. then the absolute error in the reading is 0.005mm

اقل تدريج في هذه الاداة هو 0.01mm لذا يكون الخطأ في هذه الاداه هو 0.005mm



$$12.5\text{mm} + 0.16\text{mm} = (12.660 \pm 0.005)\text{mm}$$

# Procedure

- Use the Vernier caliper to measure the diameter (**d**) of the five different cylinders.
- Use the Micrometer to measure the height (**h**) of each cylinder.
- Calculate the volume (**V**) of the cylinder.
- Weight the cylinders to find the mass (**m**) of each cylinder.
- Tabulate your readings in the table shown below.
- Plot **m** versus **V** on a graph paper and draw the best straight line through the points.
- Find the slope. From the slope calculate the density **ρ**.
- Calculate the relative and absolute errors in the density **ρ**.
- What are the possible sources of errors in this experiment?

| cylinder | Height<br>$h$ ( )<br>$\pm$ ( ) | Diameter<br>$d$ ( )<br>$\pm$ ( ) | $d^2$<br>( ) | Volume<br>$V$ ( ) | Mass<br>$m$ ( )<br>$\pm$ ( ) |
|----------|--------------------------------|----------------------------------|--------------|-------------------|------------------------------|
| 1        |                                |                                  |              |                   |                              |
| 2        |                                |                                  |              |                   |                              |
| 3        |                                |                                  |              |                   |                              |
| 4        |                                |                                  |              |                   |                              |
| 5        |                                |                                  |              |                   |                              |

***THANK YOU***

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